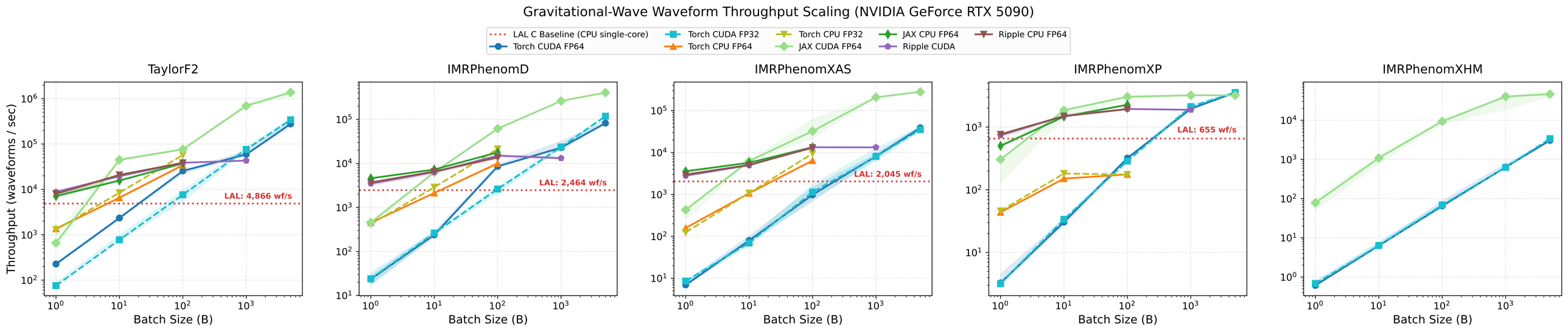
Gravitational-Wave Waveform Throughput Scaling (NVIDIA GeForce RTX 5090)
LAL C Baseline (CPU single-core)
Torch CUDA FP64
Torch CUDA FP32
Torch CPU FP64
Torch CPU FP32
JAX CUDA FP64
JAX CPU FP64
Ripple CUDA
Ripple CPU FP64
TaylorF2
IMRPhenomD
IMRPhenomXAS
IMRPhenomXP
IMRPhenomXHM
Throughput (waveforms / sec)
Batch Size (B)
LAL: 4,866 wf/s
LAL: 2,464 wf/s
LAL: 2,045 wf/s
LAL: 655 wf/s